Team 5: Adaptive Architecture for Multi-Strategy Information Retrieval

- **The Problem:** Employees spend too much time navigating policies, SOPs, and guidelines. Finding the right information can feel frustrating and inefficient.
- **The Challenge:** Traditional keyword search often misses relevant content, while cloud-based AI solutions risk exposing sensitive institutional data.
- **Our Solution:** AAMSI-R, an on-prem, privacy-safe framework that understands natural-language questions and retrieves accurate answers.
- **The Benefit:** Users get clear, grounded answers along with source documents, without ever sending sensitive data outside the organization.
- **The Impact:** Employees save time, administrators ensure compliance, and the organization can leverage its knowledge base more effectively.

Key Requirements

- **Functional (FRs)**
  - **Query Interface Dashboard:** A web-based dashboard where users can ask natural-language queries and get answers with cited source documents.
  - **Retrieval Configuration Panel:** Settings page to select and switch between retrieval strategies (syntactic, semantic, agentic) and models.
  - **Document Management Module:** Admin-only interface to upload, view, and manage documents for ingestion and indexing.
  - **Result Exploration:** Allow users to expand answers, view source snippets, and navigate to original documents.
  - **User Feedback Module:** Enable users to rate responses (e.g., helpful / not helpful) to improve retrieval quality.

- **Non-Functional (NFRs)**
  - **Availability:** System uptime of ≥99% during working hours (6am to 10pm).
  - **Latency:** Query response time ≤15 seconds (P95) for standard queries.
  - **Throughput:** Support upto 50 concurrent users without performance degradation.
  - **Scalability:** System shall handle up to 10K indexed documents without perf. deg.
  - **Modifiability:** New retrieval strategies / models should be integrable within 100 man-hours.
  - **Data Security:** 100% of data remains on-premise, with AES-256 encryption at rest and TLS 1.2+ in transit.